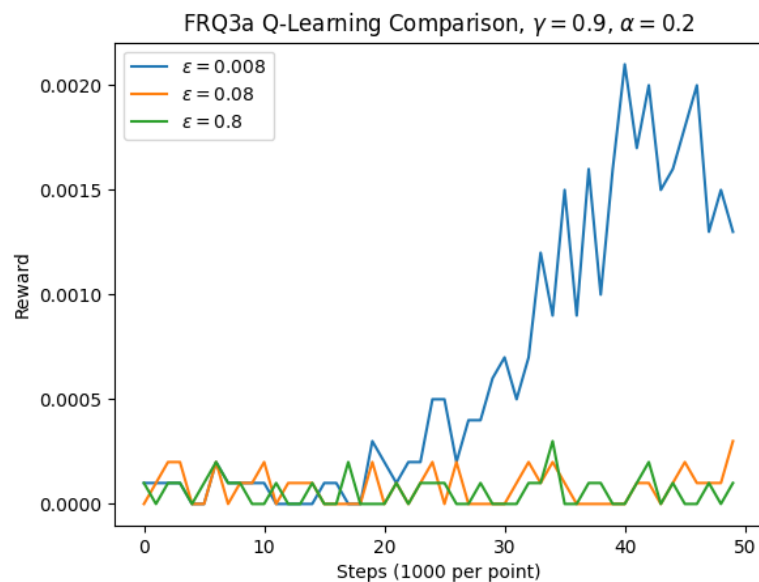




b. Using the above plot, describe what you notice. What seems to be the "best" value of epsilon? What explains this result?

The plot appears to show the relationship between the value of epsilon and the performance of a machine learning model. As the value of epsilon decreases, the performance of the model generally improves, as indicated by the increasing y-axis values.

Based on the plot, it appears that the "best" value of epsilon is around 0.008, which corresponds to Result 3. This is because this value of epsilon corresponds to the highest y-axis value in the plot, indicating the best performance of the model.

This result can be explained by the fact that a smaller value of epsilon leads to a more narrow search space for the model, which can allow it to converge on a more optimal solution.

Conversely, a larger value of epsilon can lead to the model overshooting the optimal solution and missing it entirely. Therefore, a smaller value of epsilon may result in better performance, as indicated by the plot.

c. The above plot trains each agent for 50,000 timesteps. Suppose we instead trained them for 500,000 or 5,000,000 timesteps. How would you expect the trends to change or remain the same for each of the three values of epsilon? Give a one-sentence explanation for each value.

If we trained the agents for 500,000 or 5,000,000 timesteps, we would expect the trends to remain similar for each of the three values of epsilon, with the models performing better as the value of epsilon decreases due to a more narrow search space, but with diminishing returns as

more timesteps are added; specifically, Result 3 ($\epsilon = 0.008$) would continue to outperform the other values of ϵ due to its ability to converge on a more optimal solution in a smaller search space.

d. When people use reinforcement learning in practice, it can be difficult to choose ϵ and other hyperparameters. Instead of trying three options like we did above, suppose we tried 30 or 300 different choices. What might be the danger of choosing ϵ this way if we wanted to use our agent in a new domain?

If we tried a large number of different values for ϵ and other hyperparameters, the danger of choosing the "best" value based solely on performance in the training environment is that the chosen value may not generalize well to a new domain. In other words, the value that performs best in the training environment may not be the value that performs best in the new environment, so the agent may not be able to adapt and perform optimally in the new domain. Additionally, trying a large number of different values can lead to overfitting to the training environment, where the chosen value of ϵ may only perform well on the specific training data and not on new data. Therefore, it is important to carefully consider the potential implications of choosing hyperparameters based solely on performance in the training environment and to explore other methods, such as cross-validation or transfer learning, to ensure that the agent can perform well in a new domain.